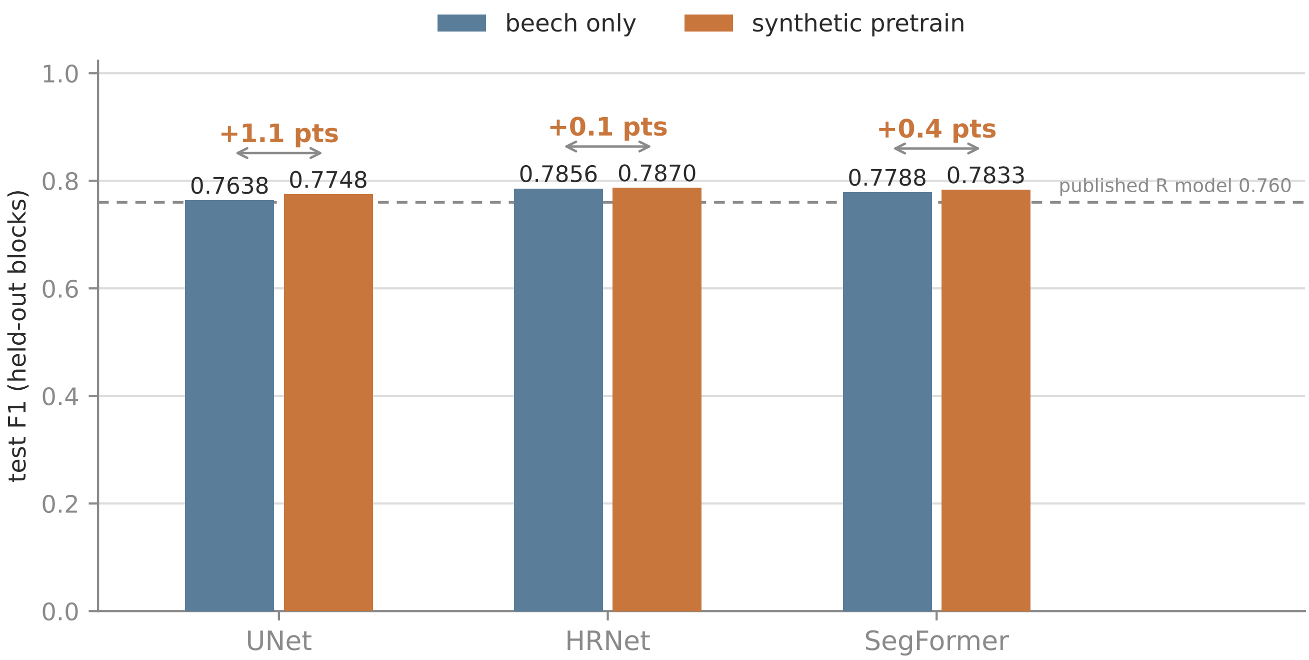
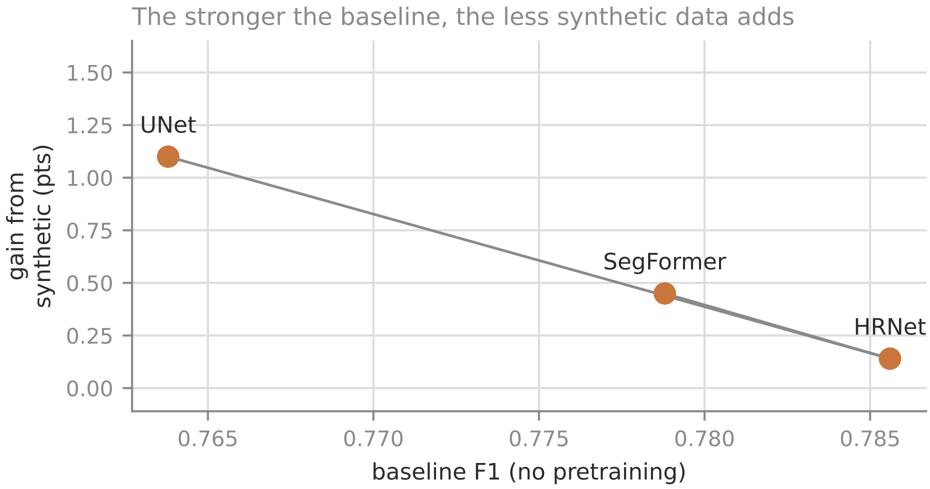
Does synthetic pretraining help a beech stem segmenter?

Yes — but less the better the architecture. Leak-free block split, one run per arm.





The gain is not a constant. It shrinks monotonically as the architecture strengthens — and HRNet's baseline, with no pretraining at all, already beats UNet's pretrained arm.

So synthetic data substitutes for model capacity here rather than adding what a stronger model lacks. On this corpus, changing architecture buys more than pretraining does, and the two do not stack.

One run per arm: HRNet's +0.1 is indistinguishable from seed noise. The architecture ranking is a larger effect and more likely real.

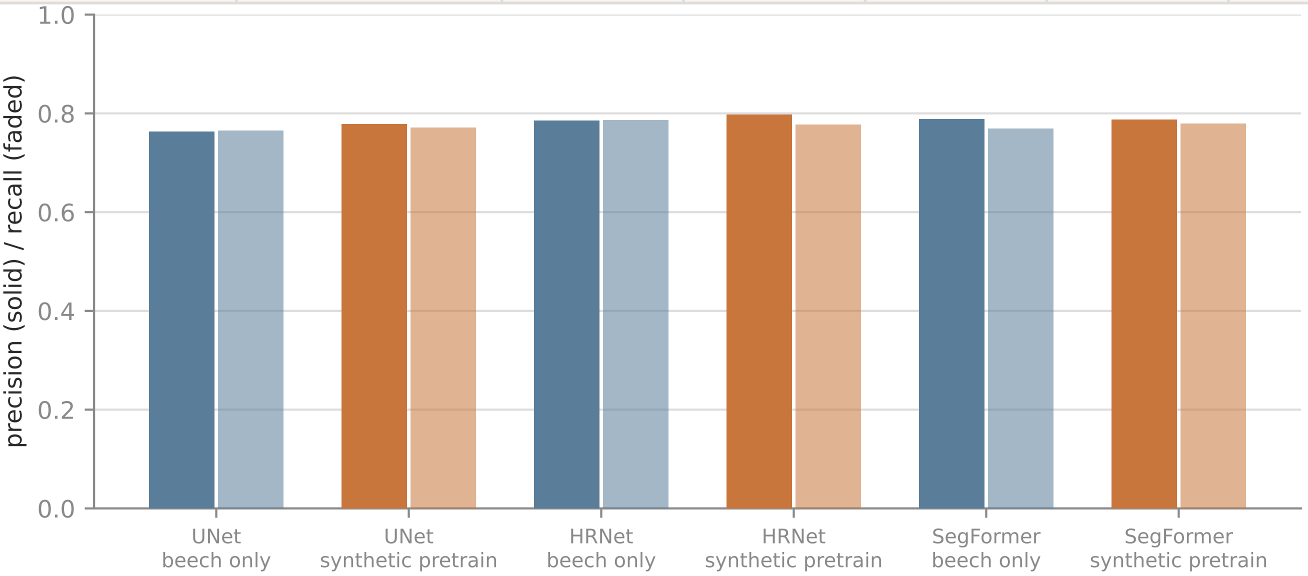
Why this matters here

The corpus holds 9,802 m² of digitized stem in total — under one hectare, across 3,842 polygons and 1,777 trees, and only 1 unannotated beech orthomosaic remains. Beech labelling is close to exhausted, so a generator producing unlimited perfectly-labelled tiles is one of the few sources of signal left — and the architecture result above says it is not the cheapest one to reach for first.

Results in full

Every value read from the run's own test_results.md

architecture	arm	test F1	precision	recall	test loss	val F1
UNet	beech only	0.7638	0.7626	0.7650	0.3418	0.7527
UNet	synth pretrain	0.7748	0.7785	0.7712	0.3276	0.8173
HRNet	beech only	0.7856	0.7852	0.7860	0.3104	—
HRNet	synth pretrain	0.7870	0.7972	0.7770	0.3124	—
SegFormer	beech only	0.7788	0.7883	0.7694	0.3217	—
SegFormer	synth pretrain	0.7833	0.7871	0.7796	0.3182	—



Setup

30 epochs, batch 16, fixed --val-data-dir and --test-data-dir so neither arm re-splits. Two-stage resets the learning rate between stages, so stage 2 does not inherit stage 1's decayed rate. The smp architectures use encoder_weights=None: an ImageNet encoder would confound what the synthetic stage buys.

Real tiles are sampled at --extent 10.24, i.e. 512 px / 10.24 m = 2 cm/px, exactly the synthetic generator's gsd_m_per_px. Sharing a ground resolution between the two stages is the point of pretraining.

train 3,084 tiles (Campus 2,500 + Kaufland 584) · val 500 · test 570 · stage 1: 3,000 synthetic tiles

Two designs that were wrong first

Both produced numbers that looked like results

1 · Holding out a whole site gave F1 exactly 0.0000 — for every arm

Peak output probability was 0.0024: the models never fired. The labels were fine — at native resolution the stems are visible and correctly aligned. The cause was colour.

set	mean RGB	amber pixels
Campus (train)	0.428 / 0.444 / 0.377	19.8%
Kaufland (train)	0.531 / 0.631 / 0.461	— (66% green)
Bachsee_north (held-out site, test)	0.478 / 0.279 / 0.126	100%
Synthetic (stage 1)	0.336 / 0.309 / 0.312	20.8%

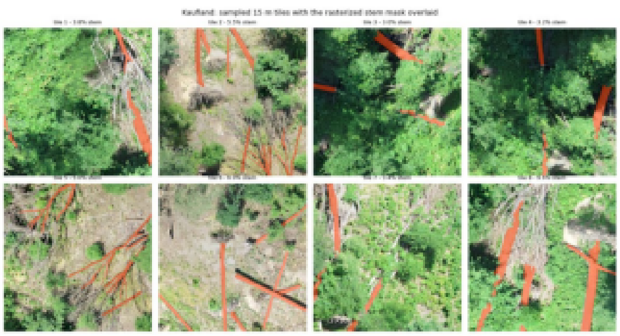
Bachsee_north is a November beech canopy in full autumn colour — a domain nothing else covers, synthetic included. With three beech sites, holding one out removes an entire acquisition: its phenology, colour cast and GSD. The score then measures domain transfer, not segmentation. Hence spatial block splits: whole blocks per split, each shrunk by the footprint half-diagonal so no tile straddles a boundary at any rotation.

2 · The stem filter tested the wrong square

Acceptance measured a world-space footprint rotated counter-clockwise, but PIL rotates the image counter-clockwise — which samples the world square rotated the other way. They coincide only at 0° and 90°. Image and mask still rotated together, so labels stayed registered and an overlay looked perfect. Only the selection was wrong: 3.2% of written tiles fell below the 0.5% stem floor and 21 were completely empty.

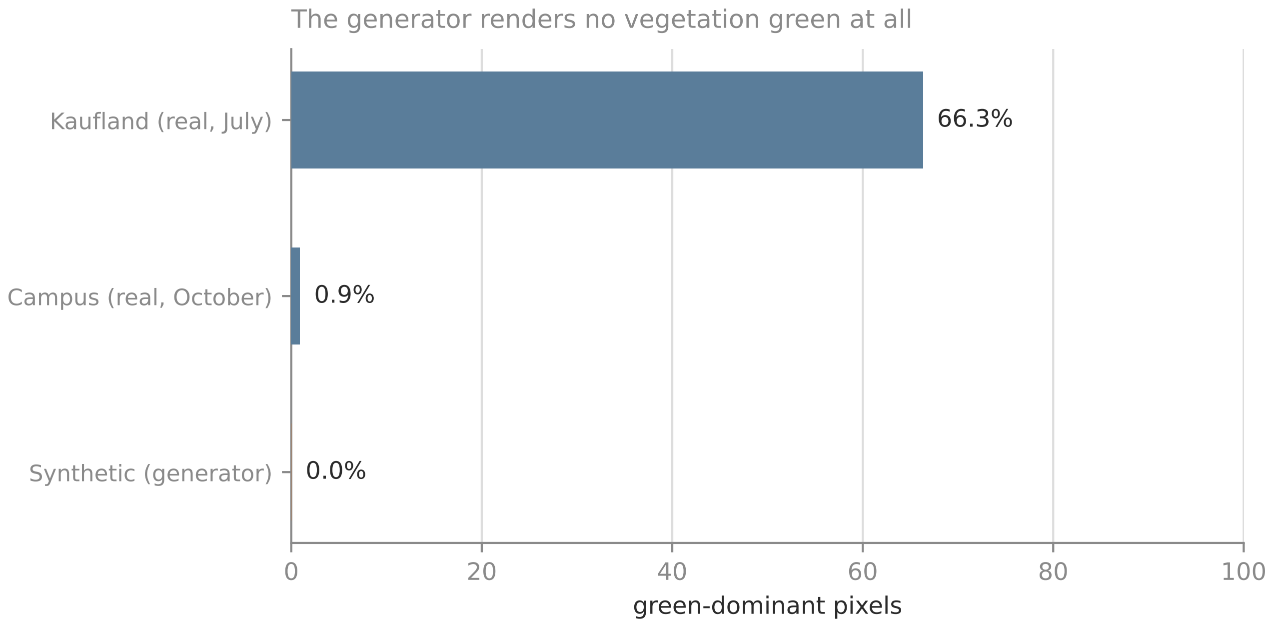
The floor now applies to the finished mask, which is exact by construction. After the fix: 0 empty, 4 of 4,154 marginally under floor. The pre-existing test could not catch it — its stem fixture is dense enough that every square contains stems whichever way it turns.

Sampled tiles with rasterized masks — the check that looked clean and hid the bug



What the synthetic data does and does not cover

Scope of the claim, and where it would not hold



The synthetic palette matches autumn and leaf-off scenes and misses leaf-on entirely. 24% of its tiles also carry >10% violet pixels — a colour no orthomosaic produces.

So the gain reported here is measured on the conditions the generator resembles. A leaf-on summer target would likely benefit less until the generator grows a summer palette. That is a scope limit on the claim, not a reason to discount it.

What probably makes it work despite the colour defects is that the label statistics line up: synthetic stem coverage averages 7.0% against 6.2% for real tiles. For a segmentation prior, the geometry of the labels matters more than the palette.

Next

- Seed replicates — run_train has no --seed CLI flag yet (it lives in TrainConfig); adding it turns the gap from a direction into a defensible effect size.
- Bachsee_north stays a legitimate hard-transfer benchmark; it is not a fair test set.
- A summer palette in the generator would extend the claim to leaf-on sites.
- For beech, the remaining levers are pre-labelling with the released ONNX plus human correction, synthetic data, and generic→species transfer — not new annotation.